

## LINCOLN 8 ENE, NE, USA

WMO: 744420

Lat: 40.8484N

Lon: 96.5651W

Elev: 1189

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94995

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -3.3       | 1.7 | -10.0                         | 3.3 | -1.5 | -5.5 | 4.3 | 3.1  | 12.5                  | 33.5 | 10.8 | 31.8 | 1.4                   | N/A  | 0.410 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.4                   | 94.3            | 74.8 | 91.4 | 74.9 | 88.7 | 73.9 | 79.0                | 89.3 | 77.2 | 87.0 | 75.5 | 85.3 | 4.1                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.1                            | 142.6 | 86.1 | 74.3 | 133.8 | 83.9 | 72.5 | 125.7 | 81.9 | 43.4          | 89.7 | 41.5 | 87.2 | 39.8 | 85.5 | 83.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |  |  |
| 10.0              | 8.3  | 6.9 | -12.2                      | 98.9 | 5.6                | 2.7 | -16.2                                              | 100.8 | -19.5      | 102.4 | -22.6      | 103.9 | -26.6      | 105.8 |  |  |
|                   |      |     | -13.3                      | 81.7 | 4.1                | 1.4 | -16.2                                              | 82.7  | -18.6      | 83.5  | -20.9      | 84.3  | -23.9      | 85.3  |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 51.5   | 24.6  | 27.4  | 41.1  | 51.9  | 62.3  | 72.2  | 76.2  | 73.6  | 66.3  | 52.9  | 39.9  | 28.2  |
|                                                                               | DBStd       | 20.60  | 12.44 | 12.41 | 12.12 | 10.33 | 8.77  | 6.31  | 6.18  | 5.92  | 8.41  | 9.87  | 10.40 | 11.04 |
|                                                                               | HDD50       | 2952   | 787   | 635   | 329   | 100   | 10    | 0     | 0     | 0     | 2     | 83    | 330   | 676   |
|                                                                               | HDD65       | 6007   | 1252  | 1052  | 742   | 408   | 159   | 12    | 3     | 6     | 85    | 393   | 754   | 1141  |
|                                                                               | CDD50       | 3503   | 0     | 3     | 54    | 156   | 390   | 667   | 811   | 732   | 491   | 172   | 26    | 1     |
|                                                                               | CDD65       | 1086   | 0     | 0     | 3     | 14    | 74    | 229   | 350   | 274   | 125   | 17    | 0     | 0     |
|                                                                               | CDH74       | 11210  | 0     | 0     | 46    | 218   | 821   | 2333  | 3722  | 2608  | 1249  | 204   | 9     | 0     |
|                                                                               | CDH80       | 4372   | 0     | 0     | 8     | 61    | 259   | 861   | 1641  | 1056  | 436   | 50    | 0     | 0     |
| Wind                                                                          | WSAvg       | 2.7    | 2.9   | 3.1   | 3.5   | 3.6   | 2.9   | 2.4   | 2.0   | 1.9   | 2.1   | 2.4   | 2.7   | 2.8   |
| Precipitation                                                                 | PrecAvg     | 30.2   | 0.7   | 0.9   | 1.5   | 2.8   | 5.0   | 4.6   | 3.8   | 3.5   | 2.8   | 2.1   | 1.5   | 1.0   |
|                                                                               | PrecMax     | 43.3   | 1.5   | 2.2   | 3.7   | 4.8   | 10.5  | 8.8   | 12.3  | 7.3   | 6.1   | 5.1   | 3.3   | 4.3   |
|                                                                               | PrecMin     | 19.5   | 0.1   | 0.1   | 0.0   | 0.6   | 1.4   | 0.4   | 0.9   | 0.4   | 0.4   | 0.1   | 0.1   | 0.0   |
|                                                                               | PrecStd     | 5.8    | 0.4   | 0.6   | 1.0   | 0.9   | 2.6   | 2.2   | 2.4   | 1.9   | 1.3   | 1.4   | 0.9   | 0.9   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 59.6  | 67.9  | 79.1  | 86.9  | 91.9  | 94.9  | 99.7  | 97.4  | 92.7  | 86.6  | 74.5  | 59.9  |
|                                                                               |             | MCWB   | 45.3  | 50.7  | 59.8  | 64.7  | 68.4  | 75.8  | 72.1  | 74.1  | 72.5  | 64.6  | 56.2  | 47.5  |
|                                                                               | 2%          | DB     | 52.3  | 57.5  | 73.1  | 80.3  | 86.2  | 91.2  | 95.0  | 93.0  | 89.0  | 79.2  | 67.2  | 53.9  |
|                                                                               |             | MCWB   | 42.3  | 47.1  | 55.6  | 61.1  | 68.0  | 75.0  | 76.1  | 74.6  | 72.4  | 62.1  | 53.4  | 45.1  |
|                                                                               | 5%          | DB     | 46.7  | 51.0  | 67.0  | 74.8  | 82.6  | 88.2  | 91.8  | 89.3  | 85.6  | 75.0  | 61.9  | 48.9  |
|                                                                               |             | MCWB   | 38.8  | 41.9  | 53.3  | 57.9  | 66.3  | 73.8  | 76.9  | 74.0  | 70.9  | 59.9  | 50.4  | 41.3  |
|                                                                               | 10%         | DB     | 41.5  | 44.9  | 61.2  | 70.1  | 78.5  | 85.2  | 88.7  | 86.0  | 81.3  | 70.0  | 56.9  | 44.0  |
|                                                                               |             | MCWB   | 36.1  | 39.1  | 50.3  | 55.7  | 64.3  | 71.7  | 75.9  | 73.1  | 68.4  | 56.9  | 47.4  | 38.3  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 45.9  | 54.5  | 63.5  | 65.9  | 73.3  | 80.0  | 81.8  | 80.3  | 76.3  | 68.8  | 59.4  | 53.3  |
|                                                                               |             | MCDB   | 57.6  | 62.9  | 76.9  | 79.5  | 84.3  | 90.4  | 91.8  | 90.5  | 88.0  | 77.3  | 66.5  | 57.3  |
|                                                                               | 2%          | WB     | 42.5  | 48.1  | 59.4  | 62.6  | 70.8  | 77.4  | 79.6  | 77.7  | 74.5  | 64.5  | 55.2  | 46.0  |
|                                                                               |             | MCDB   | 50.5  | 59.3  | 68.0  | 75.2  | 82.0  | 87.8  | 89.9  | 87.0  | 85.7  | 74.2  | 65.3  | 52.4  |
|                                                                               | 5%          | WB     | 39.4  | 42.6  | 55.2  | 60.0  | 68.9  | 75.6  | 77.9  | 75.7  | 72.7  | 61.6  | 51.5  | 42.1  |
|                                                                               |             | MCDB   | 45.3  | 50.6  | 65.9  | 70.1  | 80.3  | 85.9  | 87.7  | 84.9  | 82.1  | 70.3  | 60.4  | 49.2  |
|                                                                               | 10%         | WB     | 36.4  | 38.9  | 51.1  | 57.3  | 66.6  | 73.8  | 76.2  | 74.1  | 70.9  | 58.6  | 48.1  | 38.6  |
|                                                                               |             | MCDB   | 41.2  | 44.9  | 62.4  | 66.5  | 76.8  | 83.6  | 85.6  | 82.8  | 78.9  | 66.7  | 56.5  | 43.9  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 19.8  | 20.6  | 23.6  | 24.4  | 22.6  | 21.0  | 21.4  | 21.9  | 24.1  | 24.2  | 23.1  | 19.6  |
|                                                                               |             | MCDBR  | 27.9  | 29.7  | 32.0  | 31.7  | 27.6  | 22.8  | 23.5  | 24.8  | 25.5  | 30.0  | 30.5  | 27.2  |
|                                                                               |             | MCWBR  | 18.9  | 19.2  | 19.0  | 16.5  | 12.0  | 11.4  | 10.5  | 11.1  | 11.6  | 15.9  | 17.4  | 18.3  |
|                                                                               | 5% WB       | MCDBR  | 25.5  | 28.8  | 28.7  | 25.4  | 22.8  | 21.1  | 20.0  | 20.9  | 21.1  | 24.6  | 27.2  | 25.6  |
|                                                                               |             | MCWBR  | 17.8  | 19.0  | 16.7  | 14.6  | 11.7  | 12.1  | 10.9  | 10.9  | 10.9  | 14.7  | 17.0  | 18.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.282  | 0.307 | 0.326 | 0.383 | 0.411 | 0.419 | 0.426 | 0.422 | 0.381 | 0.333 | 0.297 | 0.281 |       |
|                                                                               | taud        | 2.468  | 2.416 | 2.409 | 2.268 | 2.261 | 2.297 | 2.277 | 2.291 | 2.356 | 2.479 | 2.505 | 2.475 |       |
|                                                                               | Ebn at Noon | 281    | 288   | 294   | 282   | 275   | 272   | 269   | 267   | 271   | 275   | 273   | 272   |       |
|                                                                               | Edh at Noon | 24     | 30    | 33    | 41    | 43    | 41    | 42    | 40    | 35    | 28    | 23    | 22    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 666    | 952   | 1242  | 1579  | 1858  | 2079  | 2076  | 1807  | 1483  | 1056  | 744   | 562   |       |
|                                                                               | RadStd      | 57     | 56    | 107   | 106   | 141   | 145   | 131   | 100   | 98    | 117   | 52    | 54    |       |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page